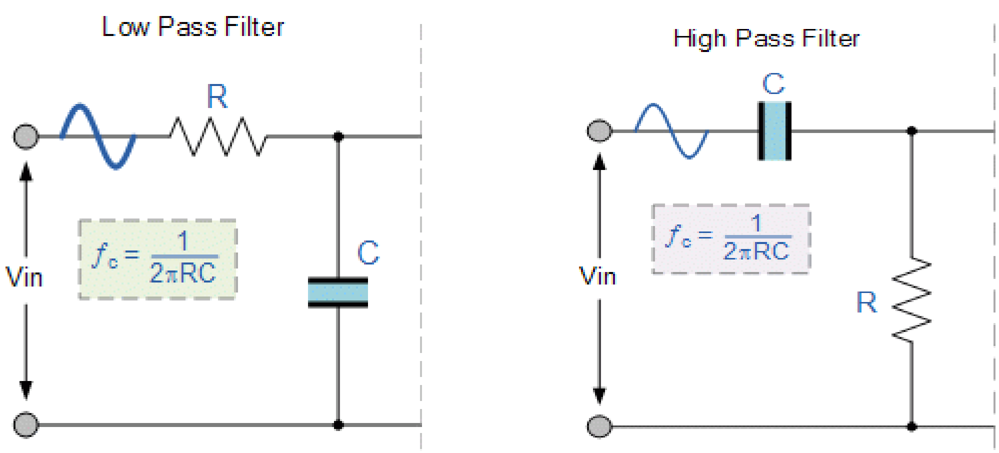
Lab 6 - Op Amp Circuits II (Bandpass Filter)

Objective

This lab is to further familiarize students with additional usages for op-amps, specifically it's usage to isolate different parts of a circuit.

Introduction

The combination of a resistor and a capacitor can be utilized to to "filter" out (attenuate) certain frequencies above, or below, the RC time constant. We'll get into more details of this over the next several chapters, but for now, we'll just introduce (without explanation) a low-pass and a high-pass filter.



In each case, the output from the filter will either attenuate frequencies above f_o (low-pass filter) or below f_o (high-pass filter). Thus, one way to create a bandpass filter is to combine these into a cascading circuit such that the low-pass filter's cutoff frequency is at the upper end of the band and the high-pass filter's cutoff frequency is at the lower end of the band.

Prelab

For the prelab, you will actually design two similar circuits in LTSpice.

Non-isolated Filters

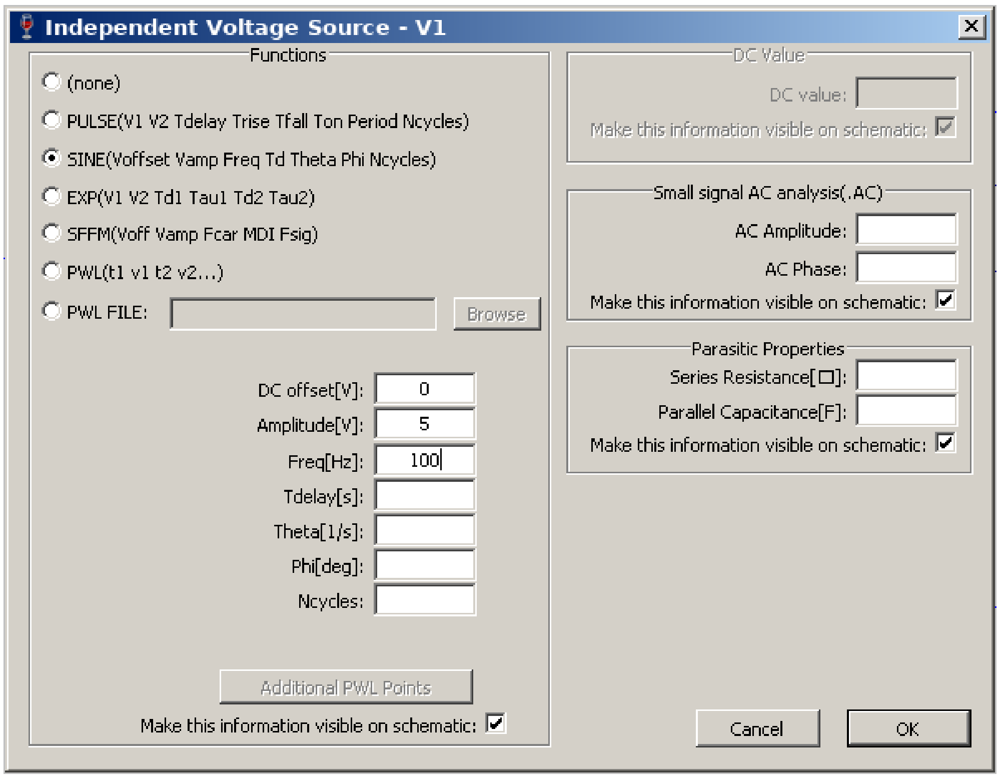
First, design the bandpass filter with just passive components (resistors and capacitors). Connect the V_{in} of the low-pass filter to a voltage source and connect the filter's output to the input of the high-pass filter. We now need to determine the values for R and C such that we attenuate all other frequencies outside of our chosen band.

For this lab, let us create a bandpass filter that allows for voice filtering - 300Hz to 3kHz. To achieve that, we need to make the low-pass filter attenuate above 3kHz and the high-pass filter attenuate below 300Hz. If we set R=10kΩ, we can calculate the C needed for each filter using:

$$C_{LP}=\frac{1}{2\pi Rf_o}=\frac{1}{2\pi(10^4\Omega)(3000\,Hz)}=5.3\,nF$$

$$C_{HP}=\frac{1}{2\pi Rf_o}=\frac{1}{2\pi(10^4\Omega)(300\,Hz)}=53\,nF$$

After putting together your circuit, you need to change the setup of the voltage source. Right click the source and choose Advanced. You will see a window pop-up:



Use the options shown, namely:

- Function = SINE
- DCoffset = 0
- Amplitude = 5
- Freq = 100 (you will adjust this to different values to see the impact to the output)

Finally, you will need to select a different SPICE command. Instead of ".op", you will need to use ".tran 10m", which tells LTSpice to simulate for 10ms and look at the transient result (we'll learn more about transient functions in Ch7). Run your simulation and look at the waveform generated at the output of the high pass filter as well as the input signal (if you click on one then the other, both waveforms will appear). Change the frequency of the SINE wave to the following values and record the max peak voltage in the following table format:

Frequency	Vout (Max)
100 Hz	
300 Hz	
500 Hz	
1 kHz	
3 kHz	
5 kHz	
10 kHz	
100 kHz	

Isolated Filters

For the second design, use the same components as before, but add two unitary gain (ideal) op-amps after each filter. To recap, you will need to take the output of the low-pass filter and feed it into the non-inverting input of the first op-amp. Take the output of this op-amp and feed it back into the inverting input as well as the Vi of the high-pass filter. Finally, attach a unitary gain op-amp to the output of the high-pass filter and run your simulation and record the same table as above, but at the output of the second op-amp. Note: don't forget to include the SPICE directive ".inc opamp.sub".

Comparison

Was there a difference between the two circuits? Which circuit operated closer to your expectation?

Procedure

Parts needed:

- Two 10kΩ resistors
- One 47nF capacitor
- One 4.7nF capacitor
- Two 741 (or equivalent) op-amps
- A selection of colored 22 gauge connection wires

It is a good idea to always verify your component values before you begin doing an experiment. This will allow you to identify one possible source of error easily.

Build and Measurements (Part 1)

- Plug in your trainer but **leave the power off**
- Build the LP and HP filters with the LP filter output connected to the HP filter input
- For the LP input, use the sine wave generator on the trainer
- **Double-check your connections!**
- Turn on power to the trainer
- Using your oscilloscope, adjust the sine wave output to be 1kHz and 5V peak-to-peak
- Connect the sine-wave output to the LP filter input
- Set your scope to show the LP filter input on CH1 and the HP filter output on CH2
- Now adjust the frequency to vary between 100Hz and 100kHz, similar to the LTSpice simulations, and record the output voltages at the various frequencies
- Note any unusual or unexpected observations

Build and Measurements (Part 2)

- Turn off the trainer
- Add the two op-amps (as unity gain) into the circuit after each filter
- Use the +12V/-12V from the PS output terminals to connect to V+/V- on the op-amps
- **Double-check your connections!**
- Turn on power to the trainer
- Once again, set your input sine wave to 5V peak-to-peak and vary the frequency between 100Hz and 100kHz, taking output measurements at the various frequencies
- Note any unusual or unexpected observations

Lab Report Requirements

- Be sure to have a title page with the usual items (e.g. lab name, your name, date).
- Procedure -- Explain what you did in your own words
- Schematic & Simulation Results – Include your LTSpice schematic and simulation results for both circuits (with and without the op-amps)
- Measured Results – Include a typed version of the measured results
- Discussion - Discuss the pros/cons of including the op-amps? Were there any oddities observed either with, or without, the op-amps? How did using 47/4.7nF caps affect the filter range?